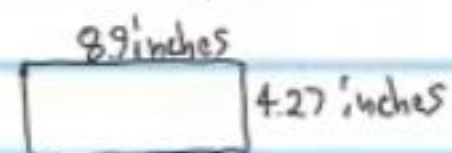


Class 36: Competitive Tests

Examples of Competitive Tests: SSAT, ISEE, COOP, HSPT, TACHS, SAT, ACT.

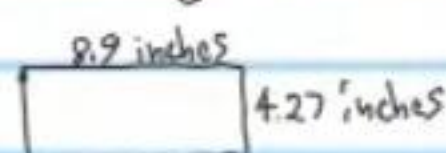
In Class Tests	VS.	Competitive Tests
Provided by your teacher.		Provided by a third party.
Designed to figure out what you know.		Designed to figure out how well you compete against other students.
Usually not multiple choice.		Almost always multiple choice.
Not a race.		Definitely a race.
Must show work.		Do not have to show work.
There shouldn't be problems that you don't recognize.		There are always problems that you won't recognize.
Not designed to exhaust you.		Designed to exhaust you.
Don't always require a great deal of effort and energy to make it through the test.		Always require a great deal of effort and energy to make it through the test.
Practice is the only way to improve your score.		Practice is the only way to improve your score.

1A What is the area of the rectangle below?



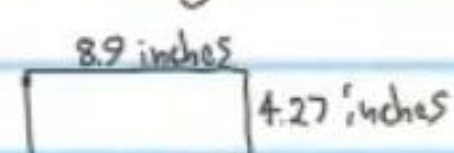
- A 100.003 in^2
- B 21.003 in^2
- C 20.003 in^2
- D 38.003 in^2
- E 13.003 in^2

1B What is the area of the rectangle below?



- A 38.004 in^2
- B 38.002 in^2
- C 38.001 in^2
- D 38.003 in^2
- E 38 in^2

1C What is the area of the rectangle below?



- A 36.003 in^2
- B 39.003 in^2
- C 37.003 in^2
- D 38.003 in^2
- E 48.003 in^2

2A If Karen bought a watch for \$17.62, a necklace for \$36.69, and a computer for \$499.76, how much did she spend in total?

- A \$515.07
- B \$615.07
- C \$600.07
- D \$690.07
- E \$554.07

2B If Karen bought a watch for \$17.62, a necklace for \$36.69, and a computer for \$499.76, how much did she spend in total?

- A \$534.07
- B \$544.07
- C \$574.07
- D \$564.07
- E \$554.07

2C If Karen bought a watch for \$17.62, a necklace for \$36.69, and a computer for \$499.76, how much did she spend in total?

- A \$554.08
- B \$554.06
- C \$554.03
- D \$554.10
- E \$554.07

3A What is the average of the following three fractions?

$$\frac{7}{8}, \frac{1}{2}, \frac{5}{8}$$

- A 0.766
- B 0.87
- C 0.57
- D 0.6
- E 0.716

3B What is the average of the following three fractions?

$$\frac{7}{8}, \frac{1}{2}, \frac{5}{8}$$

- A $\frac{3}{5}$
- B $\frac{5}{6}$
- C $\frac{17}{24}$
- D $\frac{2}{3}$
- E $\frac{19}{24}$

3C What is the average of the following three fractions?

$$\frac{7}{8}, \frac{1}{2}, \frac{5}{8}$$

- A 0.02
- B 2
- C 0.2
- D $\frac{2}{3}$
- E $1\frac{1}{8}$

4A Convert $\frac{3}{7}$ to a percent.

- A $42.\overline{857142} \%$
- B $41.\overline{857142} \%$
- C $44.\overline{857142} \%$
- D $40.\overline{857142} \%$
- E $43.\overline{857142} \%$

4B Convert $\frac{3}{7}$ to a percent.

- A $42.\overline{857142} \%$
- B $99.\overline{987} \%$
- C $178.\overline{73} \%$
- D $73.\overline{73} \%$
- E $3.\overline{73} \%$

4C Convert $\frac{3}{7}$ to a percent.

- A $42.\overline{857142} \%$
- B $42.\overline{85714} \%$
- C $42.\overline{857141} \%$
- D $42.\overline{857143} \%$
- F $42.\overline{857145} \%$

- 5A A theme park is 846 miles from the ocean, whereas a car dealership is 563 miles from the ocean. How much closer is the dealership to the ocean than the theme park?
- 5B A theme park is 846 miles from the ocean, whereas a car dealership is 563 miles from the ocean. How much closer is the dealership to the ocean than the theme park?
- 5C A theme park is 846 miles from the ocean, whereas a car dealership is 563 miles from the ocean. How much closer is the dealership to the ocean than the theme park?

A 193 mi
B 283 mi
C 203 mi
D 183 mi
E 173 mi

A 285 mi
B 283 mi
C 284 mi
D 286 mi
E 287 mi

A 83 mi
B 283 mi
C 683 mi
D 1,483 mi
E 2,483 mi

- 6A During a race, a bicyclist rides 8 miles per day. If the bicyclist rides a total of 422.4 miles during the race, for how many days was the bicyclist riding?

A 52.9
B 52.8
C 52.7
D 52.6
E 52.5

- 6B During a race, a bicyclist rides 8 miles per day. If the bicyclist rides a total of 422.4 miles during the race, for how many days was the bicyclist riding?

A 60.8
B 52.8
C 70.8
D 49.9
E 39.9

- 6C During a race, a bicyclist rides 8 miles per day. If the bicyclist rides a total of 422.4 miles during the race, for how many days was the bicyclist riding?

A 100.8
B 52.8
C 10.8
D 522.8
E 1.8

Plug in Problems

- 7 Find the missing number in the following equation.

$$(4 \times 5 \div ?)^2 + 1 = 5$$

A 2
B 5
C 4
D 10
E 20

- 8 Find z in the following equation.

$$z \times (1+1)^2 - 8 = 12$$

A 12
B 5
C 6
D 7
E 8

- 9 Eighteen divided by a number decreased by one is equal to zero. What is the number?

A 6
B 3
C 9
D 2
E 18

- 10 Five times a number increased by seven is equal to 32. What is the number?

A 3
B 2
C 5
D 6
E 4

- 11 The volume of a rectangular prism is equal to the volume of a cube. The side length of the cube is twice the width of the rectangular prism. If the height and length of the rectangular prism are 8 feet and 4 feet respectively, what is the side length of the cube?

A 4 ft
B 2 ft
C 1 ft
D 3 ft
E 5 ft

- 12 The length of a rectangle is three times the side length of a square. If the areas of the rectangle and square are equal, and the width of the rectangle is 3 inches, what is the side length of the square?

A 10 in
B 12 in
C 9 in
D 20 in
E 8 in

⑬ When a number is divided by six, the remainder is four. What could be the number?

- A 89
- B 111
- C 66
- D 79
- E 52

⑮ L is divisible by six. It's also divisible by three. Which of the following could be L?

- A 45
- B 28
- C 18
- D 15
- E 9

⑰ A student scored 88 and 85 on his first two math exams. What does he need to score on his next exam to raise his average score to 90?

- A 97
- B 98
- C 99
- D 100
- E 101

⑭ When a number is divided by three, the remainder is one. What could be the number?

- A 8
- B 9
- C 20
- D 28
- E 23

⑯ Y is divisible by 4 and 7. Which of the following could be Y?

- A 7
- B 4
- C 14
- D 56
- E 27

⑱ A student scored 10, 5, and 6 on his first three history quizzes. What does he need to score on his fourth quiz to raise his average score to 7.5?

- A 8
- B 10
- C 9
- D 12
- E 11

Precise Definitions of Math Terms

Divisible by: If X is divisible by y , then when x is divided by y , the result is a whole number. For example, 12 is divisible by 4 because when 12 is divided by 4, the quotient is a whole number (i.e. 3).

Odd Number: A number that is not divisible by 2 (e.g. 1, 3, 5, 7, 9).

Even Number: A number divisible by 2 (e.g. 0, 2, 4, 6, 8).

Distinct Numbers: Numbers different from each other (e.g. 8 & 5).

Digit: The individual numbers that make up bigger numbers. For example, the digits of 836 are 8, 3, and 6.

Prime Number: A whole number other than one that has only two factors, one and itself (e.g. 2, 3, 5, 7, 11, 13).

Integer: A whole number (e.g. -3, -2, -1, 0, 1, 2, 3).

Consecutive Numbers: Numbers that are next to each other (e.g. 7 and 8).

Multiples of A Number: The result after you multiply the number by a whole number.

EX 1) Multiples of 7: 7, 14, 21, 28, 35, etc.

EX 2) Multiples of 5: 5, 10, 15, 20, 25, 30, etc.

Factors of a number: Whole numbers we multiply to get the number.

EX 1) The factors of 10 are 1, 2, 5, and 10.

EX 2) The factors of 11 are 1 and 11.

19) What is the average of the distinct prime factors of 28?

- A $2\frac{1}{2}$
- B $3\frac{1}{2}$
- C $3\frac{1}{3}$
- D $4\frac{1}{2}$
- E 4

21) What is the average of the first three multiples of 8?

- A 16
- B 15
- C 14
- D 13
- E 12

23) What is the sum of the factors of 16?

- A 29
- B 30
- C 31
- D 32
- E 33

25) What is the product of the digits of 721?

- A 18
- B 17
- C 16
- D 15
- E 14

20) Which of the following is not the product of two prime numbers?

- A 22
- B 15
- C 10
- D 27
- E 49

22) What is the product of the first two multiples of six?

- A 80
- B 72
- C 75
- D 70
- E 81

24) What is the average of the factors of 20?

- A 4
- B 5
- C 6
- D 7
- E 8

26) What is the sum of the digits of 3,024?

- A 5
- B 8
- C 9
- D 7
- E 10

27) The sum of two consecutive even integers is 110. Which of the following is one of these integers?

- A 38
- B 48
- C 56
- D 20
- E 24

28) The product of two consecutive odd integers is 195. Which of the following is one of these integers?

- A 17
- B 7
- C 13
- D 19
- E 25

You Must Know Common Unit Conversions

1 foot (ft) = 12 inches (in)

1 Yard (yd) = 3 feet

1 pound (lb) = 16 ounces (oz)

1 ton = 2,000 pounds

1 year = 12 months = 365 days

1 week = 7 days

1 day = 24 hours

1 hour = 60 minutes (min)

1 minute = 60 seconds (s)

1 meter (m) = 100 centimeters (cm) = 1000 millimeters (mm)

1 kilometer (km) = 1,000 meters

1 gallon (gal) = 4 quarts (qt)

1 quart = 4 cups (cups)

1 dozen = 12

29) Which inequality below is correct?

- A 20 days $<$ 400 hours
- B 25 inches $<$ $1\frac{1}{2}$ feet
- C 1,000 centimeters $<$ 8 meters
- D 48 donuts $>$ 12 dozen donuts
- E $7\frac{1}{3}$ yards $>$ 21 feet

30) Which inequality below is correct?

- A 7 cups $>$ 2 quarts
- B 8 kilometers $<$ 8,000 meters
- C 5 pounds $<$ 70 ounces
- D 100,000 ounces $<$ 3 tons
- E 6,000 seconds $<$ 6 days

31) Simplify the ratio 8 feet to 6 yards.

- A $\frac{4}{3}$
- B $\frac{1}{8}$
- C $\frac{4}{9}$
- D $\frac{3}{8}$
- E $\frac{8}{18}$

32) Simplify the ratio 10 quarts to 2 gallons.

- A 3:5
- B 5:3
- C 4:5
- D 4:3
- E 5:4

33) Calculate

8 pounds 12 ounces $\div$ 1 pound 4 ounces

- A 3
- B 4
- C 5
- D 6
- E 7

34) Calculate

5 gallons 2 quarts + 4 gallons 3 quarts.

- A 14 gallons
- B 9 gallons 2 quarts
- C 9 gallons 1 quart
- D 10 gallons 1 quart
- E 10 gallons